

Online Laboratory Access Control With Zero Trust Approach: Twingate Use Case

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Abstract— As educational institutions increasingly integrate remote (online) laboratories into their systems, the need for secure and efficient access to lab resources becomes paramount. While traditional Virtual Private Networks (VPNs) have historically served as the primary solution for remote access, they are not without limitations. In this paper, we explore how zero-trust networking access solutions (ZTNA) effectively address these limitations. We examine the advantages of ZTNA-based solutions compared to traditional VPNs, discuss the underlying principles of zero-trust, and illustrate a practical application of ZTNA in educational online labs through the utilization of Twingate, a novel ZTNA-based commercial solution.

Keywords—Remote access, Online lab, Zero-Trust, ZTA, ZTNA, Twingate.

I. INTRODUCTION

In education, especially within engineering and science fields, experimental laboratories are important as they provide students with hands-on experience, allowing them to bridge the gap between theoretical knowledge and practical application. Conventionally, students in these fields have relied on physical laboratories, engaging directly with materials, and manipulating equipment to conduct experiments.

However, while physical labs remain fundamental, the adoption of online (remote or virtual) labs has become increasingly prevalent in recent years[1]. These alternative labs, supplement or replace traditional labs and enable experiments to be conducted from a distance[2]. This significance is twofold: First, it aligns with the emerging Education 4.0 paradigm, which leverages ICT advancements to provide learning experiences that transcend time and space constraints[3]. Second, this paradigm shift was further accelerated by the unprecedented challenges posed by the most recent pandemic, COVID-19. The measures imposed during the pandemic (lockdowns, social distancing) disrupted traditional modes of operation, compelling educational institutions to adapt to remote learning. Thus, the imperative for remote access to laboratory infrastructure became not only a pedagogical

enhancement but a necessity in ensuring continuous and accessible educational experiences.

To provide clarity, it is important to note that within the current scholarly discourse, researchers employ interchangeable terms that often refer to the same underlying ideas of online laboratories. The most common of these terms include remote labs, e-labs, web-labs, and tele-labs[4]. However, the IEEE Standard for Networked Smart Learning Objects for Online Laboratories (IEEE Std 1876-2019) introduced a paradigm to standardize online labs as Laboratory as a Service (LaasS)[5]. LaasS, an acronym derived from the broader “XaaS” nomenclature, follows the format where “X” signifies a variable concept and “aaS” denotes “as a Service”. This standard defines an online laboratory as “a laboratory that is accessible via computer networks, such as the Internet”. It can either be virtual, remote, or a hybrid of the two. It defines as well remote laboratory as “a set of network-connected physical devices that can be observed and controlled at distance”[5]. For the rest of this paper, the term “online labs” will be used.

While they are invaluable in their contributions to hands-on learning, online labs are not without their limitations. Larbaoui et. al [6] assert that web services of online laboratories and e-learning platforms are susceptible to vulnerabilities commonly exploited in web applications. The vulnerabilities that enable commonly known attacks such as XPath Injections and SQL Injections are recognized as prevalent and highly consequential within web services [7], [8]. These vulnerabilities pose significant risks, as they can be exploited by malicious actors to gain unauthorized access to sensitive data, manipulate or delete data, and execute arbitrary code on the lab server. The unauthorized manipulation or deletion of data can disrupt laboratory experiments, affecting the integrity of experiments and research outcomes.

Additionally, the study in [6] highlights that online lab systems are often deployed with misconfigurations that can expose them to Man-In-The-Middle (MITM) attacks. MITM attacks pose multifaceted risks: for instance, they can jeopardize network traffic, potentially compromising sensitive user information like login credentials and

personal data. Secondly, compromised credentials can enable unauthorized access to laboratory resources, allowing manipulation of experimental setups, and compromising safety and experiment integrity. MITM attacks can also disrupt online laboratory services through Denial-of-Service (DoS) or Distributed Denial-of-Service (DDoS) tactics[9], causing significant disruption to ongoing research or experiments.

The exposure of remote services stands as one of the most prevalent attack vectors, even in industrial settings[10]. If a remote service has a security flaw (e.g. misconfigurations, weak credentials), attackers can exploit it to gain unauthorized access. Virtual Private Network (VPN) solutions have been widely used in providing secure remote access for decades. However, these same VPN technologies, once considered essential for remote access, are now increasingly criticized as inadequate.

VPNs rely on a network-perimeter-based security model to provide users with full access to the internal network once authenticated. This approach increases the risk of cyberattacks, especially if malicious actors gain access to login credentials. Threat actors have increasingly targeted VPNs, with almost 760 (by the time of writing) VPN-related vulnerabilities listed in the Common Vulnerabilities and Exposures (CVE) database[11].

According to the findings presented in [12], potential cyber threats that could exploit VPN vulnerabilities include ransomware attacks, phishing attacks, malware infections, credential theft attacks, zero-day exploits, MITM attacks, and DDoS attacks, among others. Therefore, VPN-based technologies are proving inadequate in addressing modern threats, particularly as institutions transition to cloud-based solutions[13] [14].

Consequently, there has been an increasing focus on transitioning towards alternative solutions based on the Zero Trust (ZT) model in recent years[12]. Unlike conventional VPNs which grant access to the entire network and resources, Zero Trust Network Access (ZTNA) operates at a more granular level. It challenges the traditional security model by replacing the concept of a single network perimeter with micro perimeters around individual resources and requires continuous authentication, authorization, and monitoring of users, devices, and applications for every network access attempt[15].

However, during our research, we have observed that the current scholarly discourse tends to focus more on Zero Trust (ZT) as a broader and more mature concept, while ZTNA, an emerging subset of ZT that addresses the shortcomings of traditional remote access solutions has received relatively less attention. Instead, ZTNA is more prominently discussed among practitioners, as evidenced by the prevalence of grey literature (industry reports, white papers, and technical documentation) on the topic compared to academic literature. Therefore, in this paper, we discuss the underlying principles of zero-trust, examine the advantages of ZTNA-based solutions compared to traditional VPNs, and illustrate a practical application of ZTNA in educational online labs by leveraging Twingate[16], a novel ZTNA-based commercial solution.

The rest of this paper is structured as follows: Section 2 offers an overview of the zero-trust concept. Section 3 delves into the practical use case of ZTNA-based solutions.

Section 4 examines the implications of the showcased ZTNA solution, particularly in educational settings. Finally, Section 5 presents the conclusion.

II. ZERO TRUST CONCEPT- A BRIEF OVERVIEW

A. Zero Trust

Over the past decades, the traditional perimeter-based security model has come under increasing scrutiny for its inadequacy in addressing the dynamic and sophisticated nature of modern threats. Often referred to as the “castle-and-moat” security model[17], it relied on the assumption that entities within the network perimeter were inherently trusted, while those outside were considered untrusted[18]. This model proved insufficient in addressing the dynamic and distributed nature of modern enterprise networks, which are no longer confined within premises but also encompass trends like remote users, bring your own device (BYOD), and cloud-based assets that extend beyond traditional network perimeter[14], [19]. With this approach, once attackers breach the perimeter defenses, they encounter minimal resistance to lateral movement within the network[15]. Thus, it fails to address the reality of contemporary cybersecurity threats, which include both internal and external adversaries, as underscored by the increasing prevalence of insider or insider-enabled security breaches[20].

Given these challenges, a new paradigm that prioritizes proactive measures and stringent access controls over implicit trust has emerged as imperative. Consequently, the Zero Trust (ZT) security model was introduced. Although the concept has been present in cybersecurity discourse before, it was not until a principal analyst at Forrester, John Kindervag, coined and popularized the term “Zero Trust” in 2010 that the concept gained significant traction[15]. In their analysis[20], they argued against the conventional perimeter-based security model, advocating for a “never trust, always verify” approach to security. To recognize that threats can come from both external and internal sources and, therefore, emphasizes the importance of continuously verifying and validating every access request, regardless of its origin or location within the network.

At its core, ZT is not a defined technological solution but rather a comprehensive set of guiding principles that can be applied across various aspects of workflow, system design, and operations to enhance cybersecurity posture[15]. The core principles of ZT dictate that i) all entities are considered untrusted by default, ii) access is granted based on least privilege, and iii) comprehensive security monitoring is implemented[21]. With this approach, trust is earned through ongoing scrutiny, rather than assumed by default.

Google’s BeyondCorp is often cited as one of the earliest implementations of ZT-based solutions [22]–[25]. Introduced in 2011, BeyondCorp revolutionized Google’s internal security architecture by shifting away from the traditional VPN-based perimeter security model. Instead, it established a system where access decisions are determined by factors such as device and user identity, as well as the context of the request, rather than relying solely on network location[26], [27]. Subsequently, ZT has evolved and adapted with the times, gradually gaining traction across various industries as organizations increasingly recognize the limitations of perimeter-based security approaches in the

face of evolving cyber threats[15]. Presently, organizations are embracing ZT principles, whether through the adoption of aligned technologies or the gradual transition of their security strategies to embrace the ZT mindset[12], [28]–[31].

B. Zero Trust Architecture

While ZT represents a broader security philosophy that challenges the traditional assumption of implicit trust within networks, Zero Trust Architecture (ZTA) is an implementation framework that operationalizes the principles of ZT within enterprise environments. The NIST's ZTA framework, introduced in 2020, proposes the basic ZTA logical components, components relationship, workflow planning, and access policies[15].

As illustrated in Fig. 1, in addition to users and resources, ZTA comprises two core components: 1) Policy Decision Point (PDP): This centralized decision-making component evaluates access requests. It consists of two subcomponents: a) Policy Engine (PE), responsible for making access decisions based on established policies. b) Policy Administrator (PA), tasked with establishing or terminating communication paths between subjects and resources, generating session-specific authentication credentials, and executing decisions made by the policy engine. 2) Policy Enforcement Point (PEP): This component enforces security policies at various network points. It enables, monitors, and terminates connections between subjects and resources, while also communicating with the policy administrator to forward requests and receive policy updates.

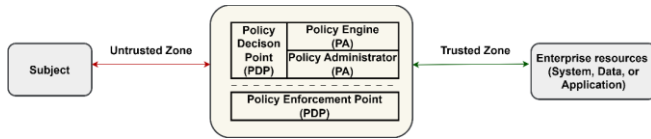


Fig. 1. Zero Trust Access

However, in addition to the core components of ZT, various enterprise data sources (both internal and external) can contribute to input and policy rules utilized by the policy engine for access decisions. As depicted by Fig. 2, these data sources can include continuous diagnostic and mitigation (CDM) systems, data access policies, system and network activity logs, industry compliance systems, threat intelligence feeds, identity management systems, security information, and event management (SIEM) system, and enterprise public key infrastructure (PKI).

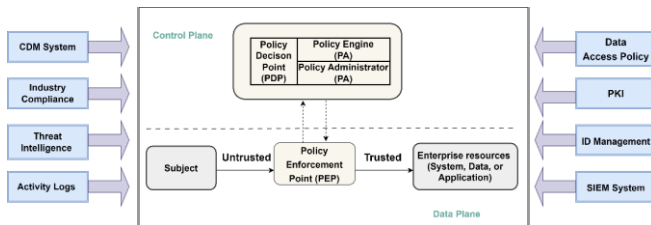


Fig. 2. Zero Trust Architecture[15]

C. Zero Trust Network Access

While ZT and ZTA represent a philosophy and framework, respectively, ZTNA encompasses both a strategic approach to the ZT security model and a range of tools and technologies that enable its implementation[32].

ZTNA has emerged as an approach to secure remote access to institutional resources and applications without relying on traditional VPNs. Introduced as part of Secure Access Service Edge (SASE), which is a collection of frameworks that unify many cloud-native cybersecurity solutions into a single concept[33], [34]. Other components of SASE focus on different areas, including software-defined wide area networking (SD-WAN), secure web gateway (SWG), cloud access security broker (CASB), and next-generation firewall (NGFW).

ZTNA-based solutions operate on a fundamentally different approach to controlling access compared to VPNs. Instead of providing broad access to the entire network, ZTNA only grants access to specific applications or resources requested by the user. This means that users are only given access to the resources they need to perform their tasks (least privileges), rather than the entire network resources. Moreover, while VPNs typically require users to authenticate once to gain access to the network, ZTNA requires continuous authentication and authorization for every access attempt. This approach ensures that access is granted on a per-session basis and is continuously monitored and verified throughout the session. Table 1 presents a concise comparison between traditional methods of securing remote access (typically VPN-based solutions) and emerging alternatives (ZTNA-based solutions).

Different tools and solutions are currently available in the market [35], offering ZTNA capabilities. These solutions typically include features such as identity-based access control, micro-segmentation, context-aware authentication, and continuous monitoring to effectively implement ZT principles in network access scenarios. In the subsequent section, we explore a practical use case by employing Twingate as one of these available solutions to establish secure remote access to our online lab. This lab is designed to facilitate student learning and experimentation with serial protocols, specifically RS232, RS422, and RS485.

III. PRACTICAL USE CASE

As mentioned earlier, this study utilizes Twingate, an alternative solution to secure remote access [36]. It differs from VPNs by adhering to the principle of least privilege, granting users access only to the resources they require following authentication and authorization.

In line with the ZT architecture, as previously discussed (Fig. 2), the Twingate controller assumes the role of a policy decision point, responsible for determining user access to resources, as illustrated in Fig. 3. The controller acts as the central coordinating element in the system. It stores configuration changes via the admin console, registers connectors, and provides signed authorizations to connecting clients. The connector functions as a policy enforcement point, serving as a centrally coordinated proxy for managing authorized traffic.

A. Remote Network

As depicted in Fig. 4, Twingate organizes resources into Remote Networks, logical containers corresponding to existing physical networks (On-premises), or Virtual Private Clouds (VPCs) intended for remote access.

Table 1. Securing remote access (Traditional Approaches vs Emerging Approaches)

Aspect	Traditional solutions (VPN-based)	Emerging Solution (ZTNA-based)
Defense Approach	— Relies on perimeter-based security. — Grants access to entire network upon login.	— Operates on a zero-trust principle. — Grants access only to specific resources requested by the user.
Security Principles	— Based on network location and trust assumptions. — Limited continuous authentication.	— Access granted on a per-session basis. — Emphasizes continuous authentication and authorization.
Scalability	— May face challenges in scaling for increasing numbers of remote users and devices.	— Generally, more scalable. — Designed to handle dynamic access requirements for distributed and remote environments. — Often leverages cloud infrastructure for scalability.
Safety Net	— Broad access may increase attack surface. — Potential for data breaches or unauthorized access.	— Granular access control minimizes attack surface. — Reduces the risk of data breaches and unauthorized access.

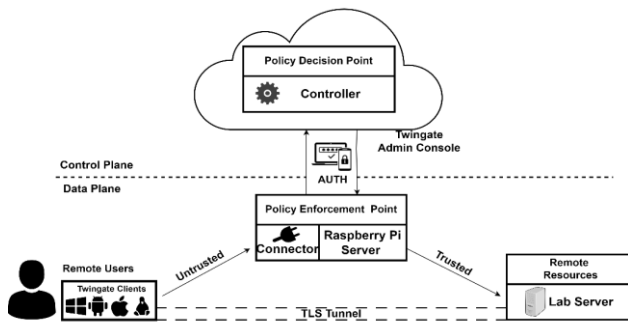


Fig.3. Deployed System Architecture

Through the Controller, administrators define the network architecture, including access rules, policies, and authentication requirements. Resources within a Remote Network must be accessible from any deployed Connector within the same network. This connector serves as the gateway for network traffic. To extend the network perimeter to remote locations, the connector establishes a secure tunnel between remote user devices and internal resources. Access to resources configured in a Remote Network depends on the presence of a deployed and operational Connector. Without a deployed and running Connector, end users cannot access the remote resources.

While there are multiple deployment options for connectors, in our scenario, connectors are employed as a "systemd" service. As illustrated in Figure 4, we deployed two connectors: the primary connector operates on a Raspberry Pi single-board computer, while an additional one operates on Kali virtual machines. Deploying multiple connectors serves to ensure load balancing, failover, and scalability.

Resources within the Twingate network are assigned specific network addresses or address ranges to be accessed through the platform as depicted in Fig.5. These resources must be reachable over the local network from the installed Connectors on the Remote Network. In our scenario, a lab server is configured as a private resource on a CIDR-defined

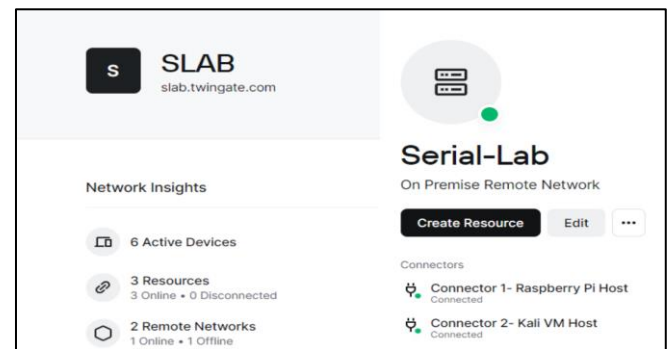


Fig.4. Deployed network for remote access to the lab resources.

subnet. The address range does not necessarily need to be routable from the end user's device. To access these resources, users are invited through the admin console via email. Upon receiving the invitation, users are provided with detailed instructions on registering for a Twingate account, granting them access to the lab server upon authentication and authorization. The lab environment is deployed on a Raspberry Pi 4B single-board computer, selected for its sufficient processing power and versatile interfaces, including USB, micro-HDMI, and Gigabit Ethernet. These features make it ideal for experimenting with serial protocols such as RS232, RS422, and RS485, as highlighted previously.

3 Resources		
Search		
Resource ↓	Address	Remote Network
Network-A	192.168.100.0/22	Serial-Lab
Raspberry Pi 4B-Lab Server	192.168.101.0/24	Serial-Lab
Windows Server	192.168.103.232	Serial-Lab

Fig.5. Deployed network resources

B. Authentication, Authorization, and Access Control

Before accessing the lab network, users are required to complete authentication and authorization processes. Access to the lab network is facilitated using a self-installed Twingate client application. To streamline the onboarding of users, Twingate seamlessly integrates with identity providers like Entra ID, Okta, OneLogin, JumpCloud, and Google. In cases where a third-party identity provider is unavailable, Twingate allows for the use of social identities for login. Currently, users have four social login options: Google, Microsoft, LinkedIn, and GitHub. Users can log in using any account from these providers. For this study, we opt for social login via Google, allowing users to authenticate without the requirement for additional login credentials (Single Sign-On).

In addition, multi-factor authentication is enforced based on user roles, specifically as a stringent requirement for individuals with admin roles, following the principles of Role-Based Access Control (RBAC). In our experiment, we opted for a Time-based One-Time Password (TOTP). Through TOTP, users can generate a unique code using a third-party authenticator app, valid for a limited duration, thereby adding a layer of security to the authentication process. As illustrated in Fig.6, after authentication, users are not granted access to the entire remote network but are restricted to specific authorized resources only. As previously highlighted, this approach ensures that users are granted access solely to the resources necessary for their tasks (least privileges). As a result, it minimizes the attack surface and mitigates the risk of lateral movement by potential attackers within the lab network.

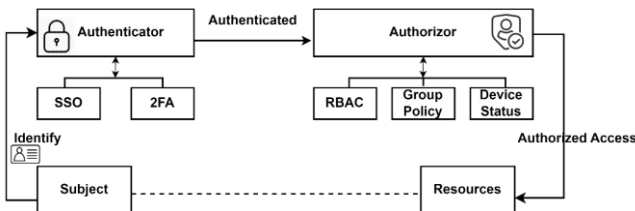


Fig.6. Authentication and authorization patterns

C. Device Security Policy

Twingate facilitates the implementation of a devices security policy tailored for remote network setups. The security policy is configured to ensure the integrity and safety of devices connecting to the network. Devices must meet the minimum OS requirements (such as hardware encryption, screen lock, firewall, antivirus) or have a trusted profile to join the deployed remote network. Hence, before granting access, devices undergo automated verification processes to confirm their identity and compliance with defined security policy. As showcased in Fig.7, access is denied for any device that does not comply with predefined policy. This approach ensures that only authorized and secure devices join the deployed remote network.

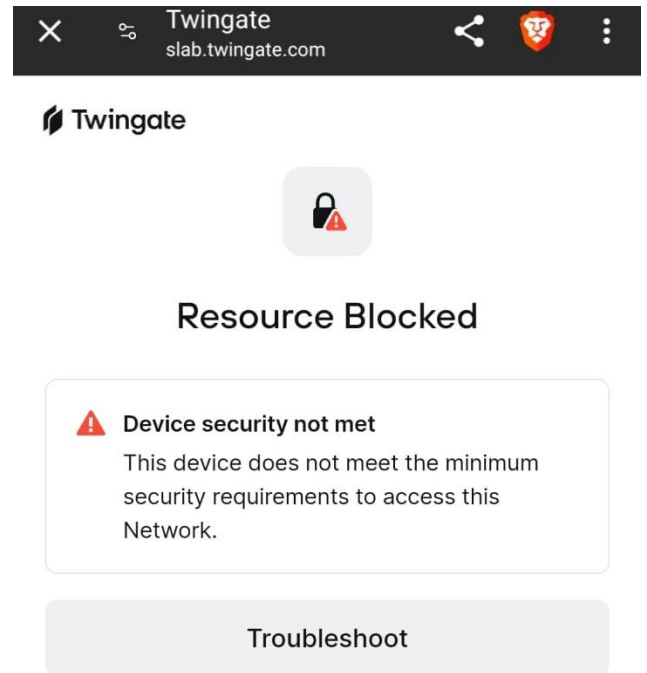


Fig.7. Device security policy enforcement

D. Encryption and Continuous Monitoring

Moreover, aside from the authentication and authorization protocols, the system mandates secure communication between clients and the connector server. This involves establishing a singular certificate-pinned Transport Layer Security (TLS) tunnel from the client to the connector, facilitated through a relay [37]. This tunnel encrypts all data-carrying traffic exchanged between the client and the connector, thereby thwarting attempts at data tampering by unauthorized entities.

Additionally, the system employs DNS-over-HTTPS (DoH) to guarantee encrypted DNS communication, safeguarding against potential DNS request manipulation by malicious actors. Furthermore, the system implements continuous monitoring and logging of all network connections established between clients and connectors as showcased by Fig.8. These logs enable administrators to promptly address any detected suspicious activity.

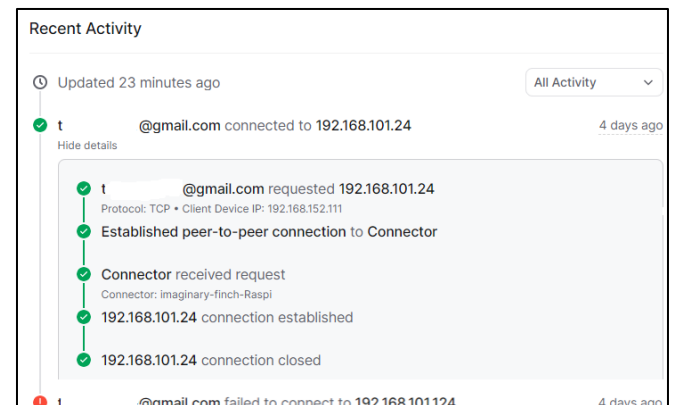


Fig.8. Activity Monitoring

IV. DISCUSSIONS

The deployed solution for managing remote access to the laboratory server network aligns closely with the principles of Zero Trust as it adheres to its foundational

tenets. The system treats all data sources and computing services as resources, requires authentication and authorization for each request, utilizes identity providers for authentication, limits user access to essential resources only, and maintains comprehensive logs of all accessed resources.

The implementation of ZTNA, exemplified by solutions like Twingate, offers significant advantages over traditional VPNs, including improved access control and enhanced security. For instance, in conventional settings, a successful attack on online lab systems can lead to lateral movement beyond lab perimeters, potentially compromising the entire institutional network. This underscores the critical importance of adopting ZTNA-based solutions in educational settings, particularly considering the increasing prevalence of cyber threats targeting educational institutions. For example, in 2022, the education sector experienced a higher proportion of ransomware victims compared to other sectors[38], indicating that ransomware attacks are no longer confined to businesses but also affect universities and colleges. Similarly, a study in 2021 [39] revealed a 75% year-on-year increase in cyberattacks targeting the education and research sector, as the transition to remote work expanded the attack surface and introduced new vulnerabilities.

Therefore, in response to evolving threats adopting ZTNA-based solutions can significantly mitigate remote exploitation risks for institutions. Fig.8 depicts an attack scenario where a breach was possible in a traditional setting (VPN-based) but prevented in a ZT-based environment.

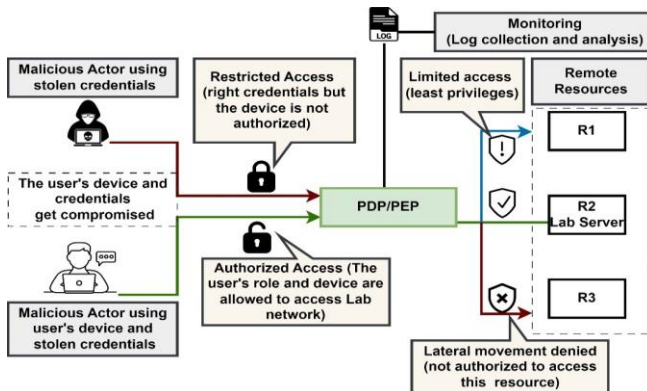


Fig.8. Attack Scenario

V. CONCLUSION

This paper has explored the concept of zero trust and its practical application through zero trust network access, with a specific focus on securing remote access to educational online laboratories. Using Twingate as an example, the study has demonstrated the practical implementation of ZTNA, showcasing how ZT principles can effectively mitigate security risks and prevent unauthorized access to laboratory resources. Future research endeavors will integrate the setup with BYOD to enhance security and manage risks associated with personal devices accessing lab resources. Furthermore, we will explore additional use cases for ZTNA in educational settings, assess the performance of different ZTNA solutions, and investigate the integration of ZTNA with other cybersecurity tools.

REFERENCES

- [1] A. Nafalski, O. Gol, Z. Nedic, J. Machotka, J. M. Ferreira, and I. Gustavsson, *Experiences with Remote Laboratories*. 2010.
- [2] R. Heradio, L. de la Torre, and S. Dormido, "Virtual and remote labs in control education: A survey," *Annu. Rev. Control*, vol. 42, pp. 1–10, Jan. 2016.
- [3] J. Miranda, C. Navarrete, J. Noguez, J.-M. Molina-Espinosa, M.-S. Ramírez-Montoya, S. A. Navarro-Tuch, M.-R. Bustamante-Bello, J.-B. Rosas-Fernández, and A. Molina, "The core components of education 4.0 in higher education: Three case studies in engineering education," *Comput. Electr. Eng.*, vol. 93, p. 107278, Jul. 2021.
- [4] M. E. Auer and C. Gravier, "Guest Editorial: The Many Facets of Remote Laboratories in Online Engineering Education," *IEEE Trans. Learn. Technol.*, vol. 2, no. 04, pp. 260–262, Oct. 2009.
- [5] "IEEE Standard for Networked Smart Learning Objects for Online Laboratories," *IEEE Std 1876-2019*, pp. 1–57, May 2019.
- [6] Y. Larbaoui, "Security, Control and Management of Smart Remote Laboratory for Remote Experiments in Electricity and Electronics," *Int. J. Adv. Trends Comput. Sci. Eng.*, vol. 9, no. 3, pp. 4068–4076, Jun. 2020.
- [7] M. Bach-Nutman, "Understanding The Top 10 OWASP Vulnerabilities."
- [8] M. A. Al-Shareeda, S. Manickam, and S. A. Sari, "A Survey of SQL Injection Attacks, Their Methods, and Prevention Techniques," in *2022 International Conference on Data Science and Intelligent Computing (ICDSIC)*, 2022, pp. 31–35.
- [9] M. Krbeček, F. Schauer, and R. Jasek, "Security aspects of remote e-laboratories," *Int J Online Eng*, vol. 9, no. 3, pp. 34–39, 2013.
- [10] "Exposed Industrial Control System Remote Services: A Threat to Critical Infrastructure | SANS Webcast." [Online]. Available: <https://www.sans.org/webcasts/exposed-industrial-control-system-remote-services-a-threat-to-critical-infrastructure/>. [Accessed: 13-Mar-2024].
- [11] "CVE - CVE." [Online]. Available: <https://cve.mitre.org/index.html>. [Accessed: 20-Mar-2024].
- [12] "2024 VPN Risk Report [HPE]," *Cybersecurity Insiders*.
- [13] "2021 VPN Risk Report - Cybersecurity Insiders | Industry Report | Zscaler." [Online]. Available: <https://info.zscaler.com/resources-industry-reports-vpn-risk-report-cybersecurity-insiders>. [Accessed: 14-Mar-2024].
- [14] C. Zanasi, S. Russo, and M. Colajanni, "Flexible zero trust architecture for the cybersecurity of industrial IoT infrastructures," *Ad Hoc Netw.*, vol. 156, p. 103414, Apr. 2024.
- [15] S. Rose, O. Borchert, S. Mitchell, and S. Connelly, "Zero Trust Architecture," National Institute of Standards and Technology, Aug. 2020.
- [16] "Twingate: It's time to ditch your VPN." [Online]. Available: <https://www.twingate.com/>. [Accessed: 30-Mar-2024].
- [17] I.-A. Dumitru, "Zero Trust Security," in *Proceedings of the International Conference on Cybersecurity and Cybercrime - 2022*, Asociatia Romana pentru Asigurarea Securitatii Informatiei, pp. 99–104.
- [18] C. Buck, C. Olenberger, A. Schweizer, F. Völter, and T. Eymann, "Never trust, always verify: A multivocal literature review on current knowledge and research gaps of zero-trust," *Comput. Secur.*, vol. 110, p. 102436, Nov. 2021.
- [19] Y. Chen, H. Hu, and G. Cheng, "Design and implementation of a novel enterprise network defense system by maneuvering multi-dimensional network properties," *Front. Inf. Technol. Electron. Eng.*, vol. 20, no. 2, pp. 238–252, Feb. 2019.
- [20] J. Kindervag, "No More Chewy Centers: The Zero Trust Model Of Information Security," 2016.
- [21] J. Kindervag, "Build Security Into Your Network's DNA: The Zero Trust Network Architecture," 2010.
- [22] H. Kang, G. Liu, Q. Wang, L. Meng, and J. Liu, "Theory and Application of Zero Trust Security: A Brief Survey," *Entropy*, vol. 25, no. 12, p. 1595, Dec. 2023.
- [23] R. Ward and B. Beyer, "A New Approach to Enterprise Security," vol. 39, no. 6, 2014.
- [24] B. Amutha, S. Saxena, and A. Das, "Changing Mechanisms of Enterprise Security (Comparing Beyond Corp with Prevalent Network Security Mechanisms)," *Int. J. Eng. Technol.*, vol. 7, no. 3.12, p. 81, Jul. 2018.
- [25] P. Assunção, "A zero trust approach to network security," in *Proceedings of the Digital Privacy and Security Conference*, 2019, vol. 2019.

- [26] "Zero Trust and BeyondCorp Google Cloud," *Google Cloud Blog*. [Online]. Available: <https://cloud.google.com/blog/topics/developers-practitioners/zero-trust-and-beyondcorp-google-cloud>. [Accessed: 15-Mar-2024].
- [27] "BeyondCorp Zero Trust Enterprise Security," *Google Cloud*. [Online]. Available: <https://cloud.google.com/beyondcorp>. [Accessed: 15-Mar-2024].
- [28] B. Pavana and S. K. Prasad, "Zero trust model: A compelling strategy to strengthen the security posture of IT organizations," *AIP Conf. Proc.*, vol. 2519, no. 1, p. 030017, Oct. 2022.
- [29] S. Ghasemshirazi, G. Shirvani, and M. A. Alipour, "Zero Trust: Applications, Challenges, and Opportunities." arXiv, 07-Sep-2023.
- [30] H. H. Nguyen, Y. Lim, M. Seo, Y. Jung, M. Kim, and W. Park, "Strengthening Information Security Through Zero Trust Architecture: A Case Study in South Korea," in *Intelligent Systems and Data Science*, Singapore, 2024, pp. 63–77.
- [31] S. Li, M. Iqbal, and N. Saxena, "Future Industry Internet of Things with Zero-trust Security," *Inf. Syst. Front.*, Mar. 2022.
- [32] "What is Zero Trust Network Access (ZTNA)?" [Online]. Available: <https://www.cloudflare.com/learning/access-management/what-is-ztna/>. [Accessed: 20-Mar-2024].
- [33] "Definition of Secure Access Service Edge (SASE) - Gartner Information Technology Glossary," *Gartner*. [Online]. Available: <https://www.gartner.com/en/information-technology/glossary/secure-access-service-edge-sase>. [Accessed: 20-Mar-2024].
- [34] M. N. Islam, R. Colomo-Palacios, and S. Chockalingam, "Secure Access Service Edge: A Multivocal Literature Review," in *2021 21st International Conference on Computational Science and Its Applications (ICCSA)*, 2021, pp. 188–194.
- [35] G. Inc, "Best Zero Trust Network Access Reviews 2024 | Gartner Peer Insights," *Gartner*. [Online]. Available: <https://www.gartner.com/market/zero-trust-network-access>. [Accessed: 21-Mar-2024].
- [36] M. Alex, "Architecting Network Connectivity for a Zero Trust Future | Twingate," 12-Jan-2022. [Online]. Available: <https://www.twingate.com/blog/network-connectivity-for-zero-trust>. [Accessed: 21-Feb-2024].
- [37] "Understanding Relays | Docs." [Online]. Available: <https://www.twingate.com/docs/understanding-relays>. [Accessed: 22-Feb-2024].
- [38] P. Mahendru, "The State of Ransomware in Education 2023," *Sophos News*, 20-Jul-2023. .
- [39] S. Shweta, "Education sector hounded by cyberattacks in 2021," *CSO Online*, 24-Jan-2022. [Online]. Available: <https://www.csoonline.com/article/571903/education-sector-hounded-by-cyberattacks-in-2021.html>. [Accessed: 23-Jan-2024].